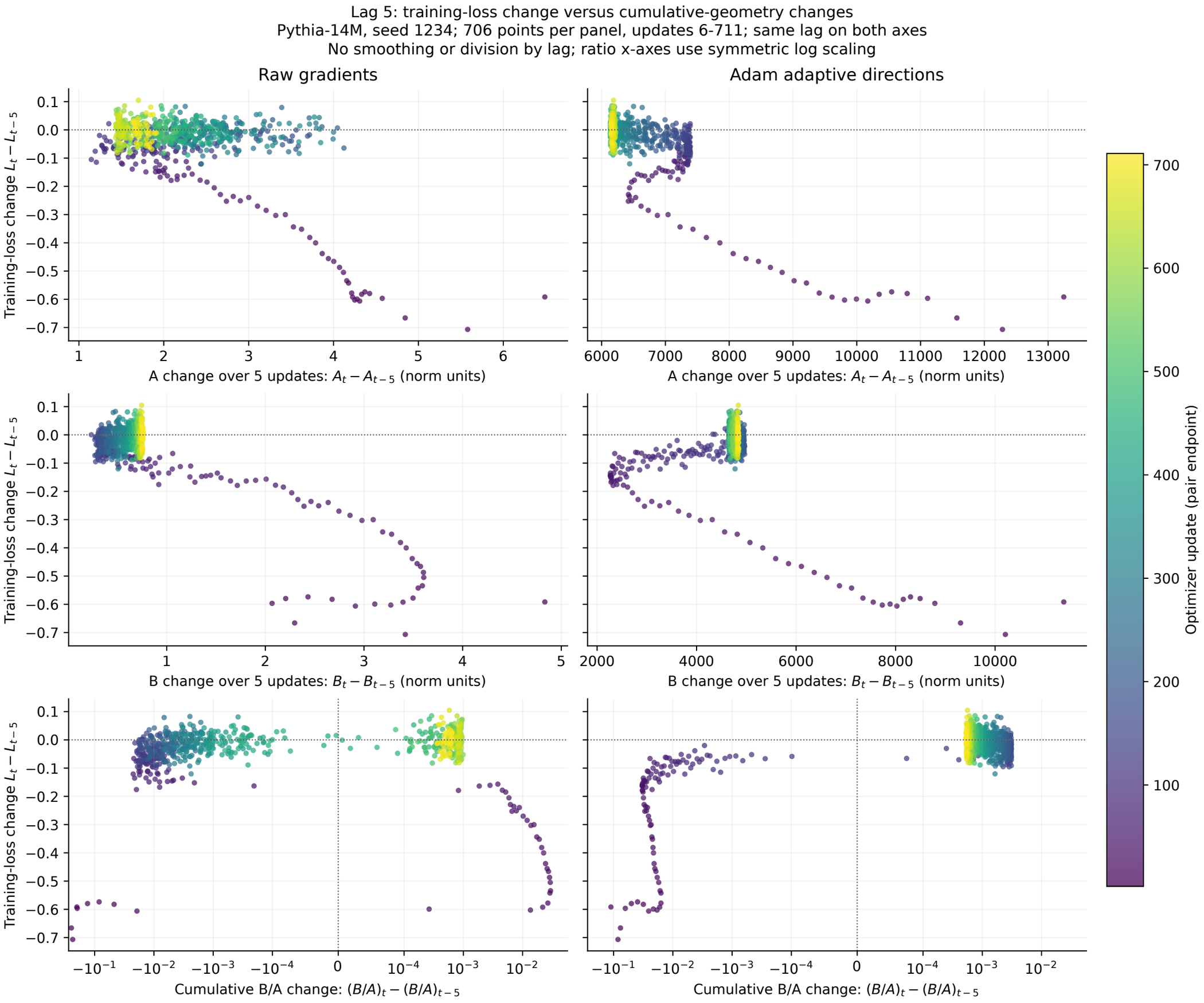

Lag 5: training-loss change versus cumulative-geometry changes
Pythia-14M, seed 1234; 706 points per panel, updates 6-711; same lag on both axes
No smoothing or division by lag; ratio x-axes use symmetric log scaling
Raw gradients
Adam adaptive directions
Training-loss change $L_t - L_{t-5}$
A change over 5 updates: $A_t - A_{t-5}$ (norm units)
B change over 5 updates: $B_t - B_{t-5}$ (norm units)
Cumulative B/A change: $(B/A)_t - (B/A)_{t-5}$
Optimizer update (pair endpoint)